

C_4	L is never eaten with C, M, or BS.	$L \cap \{C, M, BS\} = \emptyset$	Group Elimination Constraint
C_5	All 3 BS consumed before the day the 3rd O is consumed.	$Day(BS_3) < Day(O_3)$	Positional Order Constraint

4. Step-by-Step Logic Chain & Deduction Sequence

Phase 1: Determining the Fixed Positions of Flavor L

- * L must appear exactly **3 times** in the 7-day grid (Sunday through Saturday).
- * Total span required for 3 occurrences with 2-day gaps:
- * Because the week has exactly 7 available days (Positions 1 to 7), there is **zero positional elasticity**.

DEF - Zero Positional Elasticity

When the minimum required span for a constrained sequence equals the total length of the available grid, the arrangement has exactly **one valid positioning**.

Positions for L: **Sunday (Day 1), Wednesday (Day 4), Saturday (Day 7)**

Phase 2: Resolving L-Days (Sunday, Wednesday, Saturday)

- 1. Apply Elimination (C_4):** On days containing L, flavors C, M, and BS cannot appear.
- * Available candidate flavors for L-days: {L, O, M, BS, C, BC, S} minus {L, C, M, BS} = {O, BC, S}.
- 3. Evaluate Flavor O Placement on L-Days:**
- * If O were placed on an L-day, only 1 slot would remain open.
 - * The two open slots on every L-day must be filled by BC and S. Composition for Sunday, Wednesday, Saturday = **{L, BC, S}**

NOTE - Intermediate State Update

- * Flavors L, BC, and S are now completely placed (3/3 instances each).

Phase 3: Positioning Flavor BS

- * Flavors remaining to be placed: BS (3 instances), O (3 instances), M (3 instances), C (3 instances). Days with BS: **Monday, Tuesday, Thursday**

Day	Slot 1	Slot 2	Slot 3	Status
Sunday	L	BC	S	FULLY RESOLVED
Monday	BS	[]	[]	2 Slots Open ({O, M, C})
Tuesday	BS	[]	[]	2 Slots Open ({O, M, C})
Wednesday	L	BC	S	FULLY RESOLVED
Thursday	BS	[]	[]	2 Slots Open ({O, M, C})
Friday	[]	[]	[]	3 Slots Open ({O, M, C})
Saturday	L	BC	S	FULLY RESOLVED

Phase 4: Resolving Friday and Thursday

1. Friday Resolution:

* On Friday, all 3 slots are empty.

2. Thursday Resolution:

* Friday contains both **O** and **M** together. Composition for Thursday = **{BS, C, O/M}**